

The image is demonstrating **Mask R-CNN (Mask Region-Based Convolutional Neural Network)**.

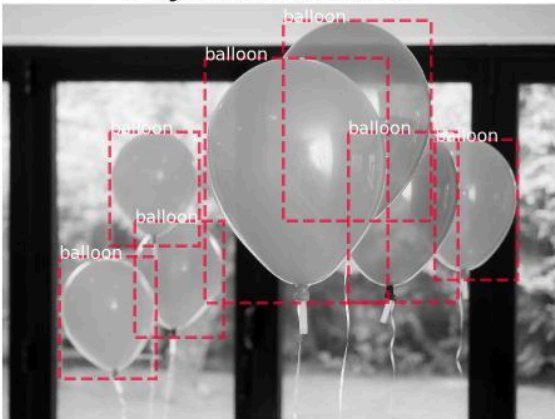
Classification



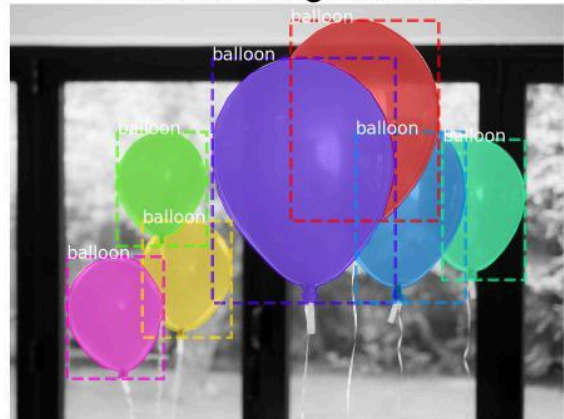
Semantic Segmentation



Object Detection

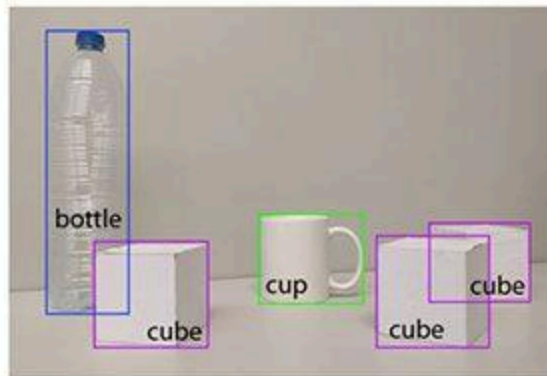


Instance Segmentation

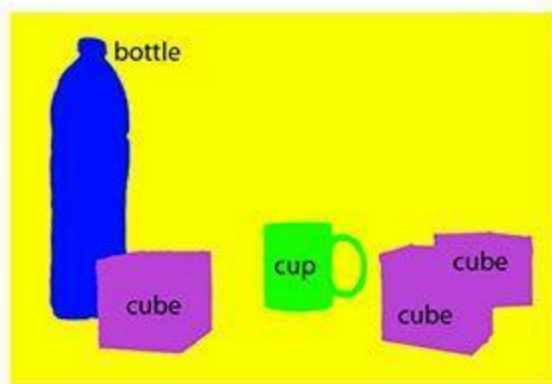




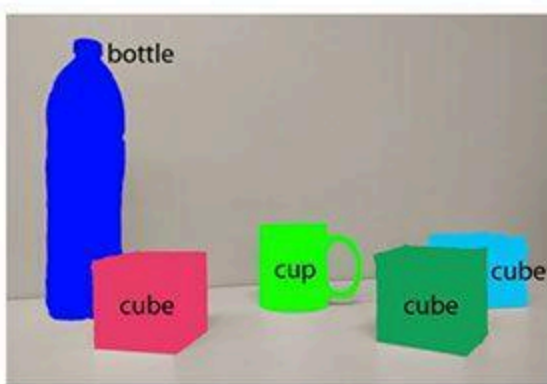
(a) Image classification



(b) Object localization



(c) Semantic segmentation



(d) Instance segmentation



What is happening in this image?

A street scene is being analyzed by an AI model.

The colored boxes and masks represent different objects detected by **Mask R-CNN**:

- 🚗 Cars
- 🚌 Buses
- 🚶 Pedestrians
- 🚲 Bicycles
- Traffic objects

Each object has:

1. A **bounding box** (rectangle around the object)
 2. A **label** (car, bus, person, etc.)
 3. A **segmentation mask** (colored region covering the exact shape)
-

Why Mask R-CNN is Special

Traditional Object Detection (YOLO)

Car → Box

Person → Box

Bus → Box

Only shows where the object is located.

Mask R-CNN

Car → Box + Exact Shape

Person → Box + Exact Shape

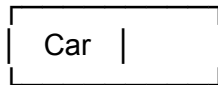
Bus → Box + Exact Shape

Shows the precise pixels belonging to each object.

Example

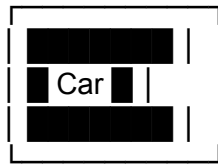
YOLO Output

[CAR]



Mask R-CNN Output

[CAR]



The green region covers only the car, not the background.

Real-Time Applications

=====

Application	Usage
Self-Driving Cars	Detect roads, cars, pedestrians
Medical Imaging	Identify tumors and organs
Video Surveillance	Track people and vehicles
Agriculture	Detect fruits, crops, weeds
Robotics	Identify objects for picking and sorting

=====

Mask R-CNN

- **Mask R-CNN** is a deep learning model used for **instance segmentation**.
- It detects objects in an image and generates a **bounding box** around each object.
- It creates a **pixel-level mask** that identifies the exact shape of the object.
- It can distinguish multiple objects even when they overlap.
- It provides both object location and object boundaries.

- It is widely used in **autonomous vehicles** for road and pedestrian detection.
- It is used in **medical imaging** to identify tumors and organs accurately.
- It is used in **surveillance systems** for people and vehicle monitoring.
- It is useful for **background removal** in photo editing applications.
- It provides more detailed information than traditional object detection models such as YOLO.

=====